The study of steady state stability is basically concerned with the determination of the upper limit of machine loadings before losing synchronism, provided the loading is increased gradually.

Dynamic instability is more probable than steady state instability. Small disturbances are continually occurring in a power system (variations in loadings, changes in turbine speeds, etc.) which are small enough not to cause the system to lose synchronism but do excite the system into the state of natural oscillations. The system is said to be dynamically stable if the oscillations do not acquire more than certain amplitude and die out quickly (i.e., the system is well-damped). In a dynamically unstable system, the oscillation amplitude is large and these persist for a long time (i.e., the system is underdamped). This kind of instability behaviour constitutes a serious threat to system security and creates very difficult operating conditions. Dynamic stability can be significantly improved through the use of power system stabilizers. Dynamic system study has to be carried out for 5–10 s and sometimes up to 30 s. Computer simulation is the only effective means of studying dynamic stability problems. The same simulation programmes are, of course, applicable to transient stability studies.

Following a sudden disturbance on a power system rotor speeds, rotor angular differences and power transfer undergo fast changes whose magnitudes are dependent upon the severity of disturbance. For a large disturbance, changes in angular differences may be so large as to cause the machines to fall out of step. This type of instability is known as transient instability and is a fast phenomenon usually occurring within 1 s for a generator close to the cause of disturbance. There is a large range of disturbances which may occur on a power system, but a fault on a heavily loaded line which requires opening the line to clear the fault is usually of greatest concern. The tripping of a loaded generator or the abrupt dropping of a large load may also cause instability.

The effect of short circuits (faults), the most severe type of disturbance to which a power system is subjected, must be determined in nearly all stability studies. During a fault, electrical power from nearby generators is reduced drastically, while power from remote generators is scarcely affected. In some cases, the system may be stable even with a sustained fault, whereas other systems will be stable only if the fault is cleared with sufficient rapidity. Whether the system is stable on occurrence of a fault depends not only on the system itself, but also on the type of fault, location of fault, rapidity of clearing and method of clearing, i.e., whether cleared by the sequential opening of two or more breakers or by simultaneous opening and whether or not the faulted line is reclosed. The transient stability limit is almost always lower than the steady state limit, but unlike the latter, it may exhibit different values depending on the nature, location and magnitude of disturbance.

Modern power systems have many interconnected generating stations, each with several generators and many loads. The machines located at any one point in a system normally act in unison. It is, therefore, common practice in stability

studies to consider all the machines at one point as one large machine. Also machines which are not separated by lines of high reactance are lumped together and considered as one equivalent machine. Thus a multimachine system can often be reduced to an equivalent few machine system. If synchronism is lost, the machines of each group stay together although they go out of step with other groups. Qualitative behaviour of machines in an actual system is usually that of a two machine system. Because of its simplicity, the two machine system is extremely useful in describing the general concepts of power system stability and the influence of various factors on stability. It will be seen in this chapter that a two machine system can be regarded as a single machine system connected to infinite system.

Stability study of a multimachine system must necessarily be carried out on a digital computer.

12.2 DYNAMICS OF A SYNCHRONOUS MACHINE

The kinetic energy of the rotor at synchronous machine is

$$\text{KE} = \frac{1}{2} J\omega_{sm}^2 \times 10^{-6} \text{ MJ}$$

where J = rotor moment of inertia in kg-m^2

ω_{sm} = synchronous speed in rad (mech)/s

But $\omega_s = \left(\frac{P}{2}\right) \omega_{sm}$ = rotor speed in rad (elect)/s

where P = number of machine poles

$$\therefore \quad \text{KE} = \frac{1}{2}\left(J\left(\frac{2}{P}\right)^2 \omega_s \times 10^{-6}\right)\omega_s$$

$$= \frac{1}{2} M \omega_s$$

where $M = J\left(\frac{2}{P}\right)^2 \omega_s \times 10^{-6}$

= moment of inertia in MJ-s/elect rad

We shall define the inertia constant H such that

$$GH = \text{KE} = \frac{1}{2} M \omega_s \text{ MJ}$$

where G = machine rating (base) in MVA (3-phase)

H = *inertia constant* in MJ/MVA or MW-s/MVA

It immediately follows that

$$M = \frac{2GH}{\omega_s} = \frac{GH}{\pi f} \text{ MJ-s/elect rad} \quad (12.1)$$

$$= \frac{GH}{180f} \text{ MJ-s/elect degree}$$

M is also called the inertia constant.

Taking G as base, the inertia constant in pu is

$$M(pu) = \frac{H}{\pi f} \text{ s}^2\text{/elect rad} \quad (12.2)$$

$$= \frac{H}{180f} \text{ s}^2\text{/elect degree}$$

The inertia constant H has a characteristic value or a range of values for each class of machines. Table 12.1 lists some typical inertia constants.

Table 12.1 Typical inertia constants of synchronous machines*

Type of Machine		*Intertia Constant H Stored Energy in MW Sec per MVA***
Turbine Generator		
Condensing	1,800 rpm	9-6
	3,000 rpm	7-4
Non-Condensing	3,000 rpm	4-3
Water wheel Generator		
Slow-speed (< 200 rpm)		2-3
High-speed (> 200 rpm)		2-4
Synchronous Condenser***		
Large		1.25
Small		1.00
Synchronous Motor with load varying from 1.0 to 5.0 and higher for heavy flywheels		2.00

It is observed from Table 12.1 that the value of H is considerably higher for steam turbogenerator than for water wheel generator. Thirty to sixty per cent of the total inertia of a steam turbogenerator unit is that of the prime mover, whereas only 4 –15% of the inertia of a hydroelectric generating unit is that of the waterwheel, including water.

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** Where range is given, the first figure applies to the smaller MVA sizes.

*** Hydrogen-Cooled, 25 per cent less.

e Swing Equation

ure 12.1 shows the torque, speed and flow of mechanical and electrical wers in a synchronous machine. It is assumed that the windage, friction and n-loss torque is negligible. The differential equation governing the rotor namics can then be written as

$$J \frac{d^2\theta_m}{dt^2} = T_m - T_e \text{ Nm} \quad (12.3)$$

here

θ_m = angle in rad (mech)

T_m = turbine torque in Nm; it acquires a negative value for a motoring machine

T_e = electromagnetic torque developed in Nm; it acquires negative value for a motoring machine

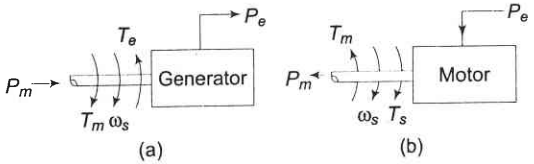


Fig. 12.1 Flow of mechanical and electrical powers in a synchronous machine

While the rotor undergoes dynamics as per Eq. (12.3), the rotor speed changes by insignificant magnitude for the time period of interest (1s) [Sec. 12.1]. Equation (12.3) can therefore be converted into its more convenient power form by assuming the rotor speed to remain constant at the synchronous peed (ω_{sm}). Multiplying both sides of Eq. (12.3) by ω_{sm}' we can write

$$J\omega_{sm} \frac{d^2\theta_m}{dt^2} \times 10^{-6} = P_m - P_e \text{ MW} \quad (12.4)$$

where

P_m = mechanical power input in MW

P_e = electrical power output in MW; stator copper loss is assumed negligible.

Rewriting Eq. (12.4)

$$\left(J\left(\frac{2}{P}\right)^2 \omega_s \times 10^{-6}\right) \frac{d^2\theta_e}{dt^2} = P_m - P_e \text{ MW}$$

where θ_e = angle in rad (elect)

or $$M \frac{d^2\theta_e}{dt^2} = P_m - P_e \quad (12.5)$$

It is more convenient to measure the angular position of the rotor with respect to a synchronously rotating frame of reference. Let

$$\delta = \theta_e - \omega_s t; \text{ rotor angular displacement from synchronously rotating reference frame (called } \textit{torque angle/power angle}) \tag{12.6}$$

From Eq. (12.6)

$$\frac{d^2\theta_e}{dt^2} = \frac{d^2\delta}{dt^2} \tag{12.7}$$

Hence Eq. (12.5) can be written in terms of δ as

$$M\frac{d^2\delta}{dt^2} = P_m - P_e \text{ MW} \tag{12.8}$$

With M as defined in Eq. (12.1), we can write

$$\frac{GH}{\pi f}\frac{d^2\delta}{dt^2} = P_m - P_e \text{ MW} \tag{12.9}$$

Dividing throughout by G, the MVA rating of the machine,

$$M(\text{pu})\frac{d^2\delta}{dt^2} = P_m - P_e;$$

in pu of machine rating as base (12.10)

where

$$M(\text{pu}) = \frac{H}{\pi f}$$

or

$$\frac{H}{\pi f}\frac{d^2\delta}{dt^2} = P_m - P_e \text{ pu} \tag{12.11}$$

This equation (Eq. (12.10)/Eq. (12.11)), is called the *swing equation* and it describes the rotor dynamics for a synchronous machine (generating/motoring). It is a second-order differential equation where the damping term (proportional to $d\delta/dt$) is absent because of the assumption of a lossless machine and the fact that the torque of *damper winding* has been ignored. This assumption leads to pessimistic results in transient stability analysis—damping helps to stabilize the system. Damping must of course be considered in a dynamic stability study. Since the electrical power P_e depends upon the sine of angle δ (see Eq. (12.29)), the swing equation is a non-linear second-order differential equation.

Multimachine System

In a multimachine system a common system base must be chosen. Let

G_{mach} = machine rating (base)

G_{system} = system base

Equation (12.11) can then be written as

$$\frac{G_{\text{mach}}}{G_{\text{system}}}\left(\frac{H_{\text{mach}}}{f}\frac{d^2\delta}{dt^2}\right) = (P_m - P_e)\frac{G_{\text{mach}}}{G_{\text{system}}}$$

or

$$\frac{H_{\text{system}}}{\pi f}\frac{d^2\delta}{dt^2} = P_m - P_e \text{ pu in system base} \tag{12.12}$$

where

$$H_{\text{system}} = H_{\text{mach}}\left(\frac{G_{\text{mach}}}{G_{\text{system}}}\right) \tag{12.13}$$

= machine inertia constant in system base

Machines Swinging Coherently

Consider the swing equations of two machines on a *common system base*.

$$\frac{H_1}{\pi f}\frac{d^2\delta_1}{dt^2} = P_{m1} - P_{e1} \text{ pu} \tag{12.14}$$

$$\frac{H_2}{\pi f}\frac{d^2\delta_2}{dt^2} = P_{m2} - P_{e2} \text{ pu} \tag{12.15}$$

Since the machine rotors swing together (coherently or in unison)

$$\delta_1 = \delta_2 = \delta$$

Adding Eqs (12.14) and (12.15)

$$\frac{H_{\text{eq}}}{\pi f}\frac{d^2\delta}{dt^2} = P_m - P_e \tag{12.16}$$

where

$$\begin{aligned} P_m &= P_{m1} + P_{m2} \\ P_e &= P_{e1} + P_{e2} \\ H_{\text{eq}} &= H_1 + H_2 \end{aligned} \tag{12.17}$$

The two machines swinging coherently are thus reduced to a single machine as in Eq. (12.16). The equivalent inertia in Eq. (12.17) can be written as

$$H_{\text{eq}} = H_{1\,\text{mach}}\, G_{1\,\text{mach}}/G_{\text{system}} + H_{2\,\text{mach}}\, G_{2\,\text{mach}}/G_{\text{system}} \tag{12.18}$$

The above results are easily extendable to any number of machines swinging coherently.

Example 12.1

A 50 Hz, four pole turbogenerator rated 100 MVA, 11 kV has an inertia constant of 8.0 MJ/MVA

(a) Find the stored energy in the rotor at synchronous speed.

(b) If the mechanical input is suddenly raised to 80 MW for an electrical load of 50 MW, find rotor acceleration, neglecting mechanical and electrical losses.

(c) If the acceleration calculated in part (b) is maintained for 10 cycles, find the change in torque angle and rotor speed in revolutions per minute at the end of this period.

Solution

(a) Stored energy = $GH = 100 \times 8 = 800$ MJ

(b) $P_a = 80 - 50 = 30$ MW $= M \dfrac{d^2\delta}{dt^2}$

$$M = \frac{GH}{180f} = \frac{800}{180 \times 50} = \frac{4}{45} \text{ MJ-s/elect deg}$$

$$\therefore \quad \frac{4}{45}\frac{d^2\delta}{dt^2} = 30$$

or

$$\alpha = \frac{d^2\delta}{dt^2} = 337.5 \text{ elect deg/s}^2$$

(c) 10 cycles = 0.2 s

Change in $\delta = \frac{1}{2}(337.5) \times (0.2)^2 = 6.75$ elect degrees

$$= 60 \times \frac{337.5}{2 \times 360^\circ} = 28.125 \text{ rpm/s}$$

$\therefore$ Rotor speed at the end of 10 cycles

$$= \frac{120 \times 50}{4} + 28.125 \times 0.2$$

$$= 1505.625 \text{ rpm}$$

12.3 POWER ANGLE EQUATION

In solving the swing equation (Eq. (12.10)), certain simplifying assumptions are usually made. These are:

1. Mechanical power input to the machine (P_m) remains constant during the period of electromechanical transient of interest. In other words, it means that the effect of the turbine governing loop is ignored being much slower than the speed of the transient. This assumption leads to pessimistic result—governing loop helps to stabilize the system.

2. Rotor speed changes are insignificant—these have already been ignored in formulating the swing equation.

3. Effect of voltage regulating loop during the transient is ignored, as a consequence the generated machine emf remains constant. This assumption also leads to pessimistic results—voltage regulator helps to stabilize the system.

Before the swing equation can be solved, it is necessary to determine the dependence of the electrical power output (P_e) upon the rotor angle.

Simplified Machine Model

For a nonsalient pole machine, the per phase induced emf-terminal voltage equation under steady conditions is

$$E = V + jX_dI_d + jX_qI_q;\ X_d > X_q \tag{12.19}$$

where
$$I = I_d + I_g \tag{12.20}$$

and usual symbols are used.

Under transient condition

$$X_d \rightarrow X'_d < X_d$$

but

$X'_q = X_q$ since the main field is on the d-axis

$X'_d < X_q$; but the difference is less than in Eq. (12.19)

Equation (12.19) during the transient modifies to

$$E' = V + jX'_dI_d + jX_qI_q \tag{12.21}$$

$$= V + jX_q(I - I_d) + jX_d'I_d$$

$$= (V + jX_qI) + j(X'_d - X_q)I_d \tag{12.22}$$

The phasor diagram corresponding to Eqs. (12.21) and (12.22) is drawn in Fig. 12.2.

Since under transient condition, $X'_d < X_d$ but X_q remains almost unaffected, it is fairly valid to assume that

$$X'_d \approx X_q \tag{12.23}$$

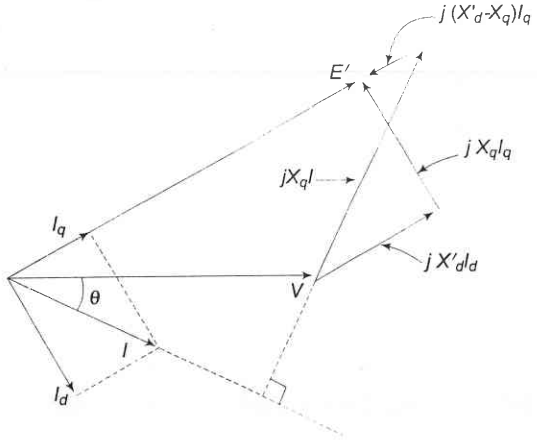


Fig. 12.2 Phasor diagram—salient pole machine

Equation (12.22) now becomes

$$E' = V + jX_qI$$

$$= V + jX_d'I \tag{12.24}$$

The machine model corresponding to Eq. (12.24) is drawn in Fig. 12.3 which also applies to a cylindrical rotor machine where $X_d' = X_q' = X_s'$(transient synchronous reactance)

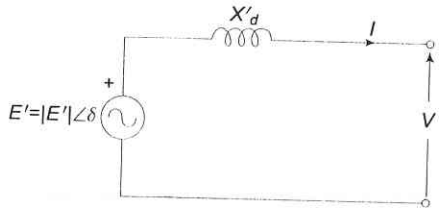


Fig. 12.3 Simplified machine model

The simplified machine of Fig. 12.3 will be used in all stability studies.

Power Angle Curve

For the purposes of stability studies $|E'|$, transient emf of generator motor, remains constant or is the independent variable determined by the voltage regulating loop but V, the generator determined terminal voltage is a dependent variable. Therefore, the nodes (buses) of the stability study network pertain to the emf terminal in the machine model as shown in Fig. 12.4, while the machine reactance (X_d') is absorbed in the system network as different from a load flow study. Further, the loads (other than large synchronous motors) will be replaced by equivalent static admittances (connected in shunt between transmission network buses and the reference bus). This is so because load voltages vary during a stability study (in a load flow study, these remain constant within a narrow band).

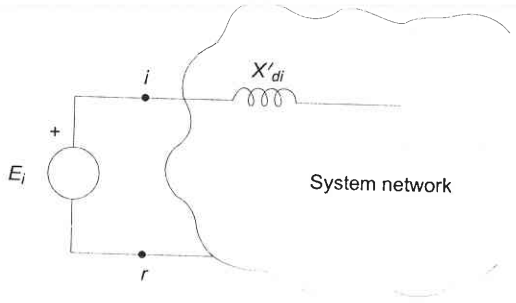


Fig. 12.4

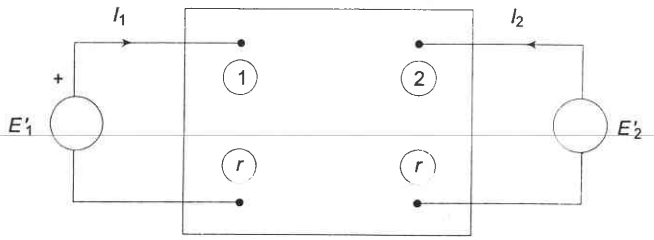


Fig. 12.5 Two-bus stability study network

For the 2-bus system of Fig. 12.5

$$\mathbf{Y_{BUS}} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}; \; Y_{12} = Y_{21} \tag{12.25}$$

Complex power into bus is given by

$$P_i + jQ_i = E_iI_i^*$$

At bus 1

$$P_1 + jQ_1 = E_1' \; (Y_{11}E_1')^* + E_1 \; (Y_{12}E_2')^* \tag{12.26}$$

But

$$E_1' = |E_1'| \angle \delta_1; \; E_2' = |E'_2| \angle \delta_2$$

$$Y_{11} = G_{11} + jB_{11}; \; Y_{12} = |Y_{12}| \angle \theta_{12}$$

Since in solution of the swing equation only real power is involved, we have from Eq. (12.26)

$$P_1 = |E_1'|^2 \; G_{11} + |E_1'| \, |E'_2| \, |Y_{12}| \cos \; (\delta_1 - \delta_2 - \theta_{12}) \tag{12.27}$$

A similar equation will hold at bus 2.

Let

$$|E_1'|^2 G_{11} = P_c$$

$$|E_1'| \, |E_2'| \, |Y_{12}| = P_{\max}$$

$$\delta_1 - \delta_2 = \delta$$

and $$\phi_{12} = \pi/2 - \gamma$$

Then Eq. (12.27) can be written as

$$P_1 = P_c + P_{\max} \sin \; (\delta - \gamma); \; \textit{Power Angle Equation} \tag{12.28}$$

For a purely reactive network

$$G_{11} = 0 \; (\because \; P_c = 0); \text{ lossless network}$$

$$\theta_{12} = \pi/2, \; \therefore \; \gamma = 0$$

Hence

$$P_e = P_{\max} \sin \; \delta \tag{12.29a}$$

Excitation States at Constant Load

- **Under-Excitation ($E_f < V$):**
  - **Synchronous Motor:** Draws a lagging current (consumes lagging reactive power), resulting in a lagging power factor. The armature current increases above its minimum value.
  - **Synchronous Generator:** Acts to supply real power and a leading power factor. YouTube · NCM learning... +1
- **Normal (Unity) Excitation ($E_f = V$):**
  - The armature current (I_a) reaches its **minimum possible value**.
  - The power factor is **unity (φ = 1)**, and zero reactive power is exchanged with the grid.
- **Over-Excitation ($E_f > V$):**
  - **Synchronous Motor:** Draws a leading current (supplies lagging reactive power / acts like a capacitor), achieving a leading power factor. This mode is commonly used for power factor correction (as a *synchronous condenser*).
  - **Synchronous Generator:** Demagnetizes the machine and alters reactive power output sharing when connected in parallel with other alternators. YouTube · Education 4u +3

POWER SYSTEM STABILITY

LESSON SUMMARY-1:-

1. Introduction
2. Classification of Power System Stability
3. Dynamic Equation of Synchronous Machine

Power system stability involves the study of the dynamics of the power system under disturbances. Power system stability implies that its ability to return to normal or stable operation after having been subjected to some form of disturbances.

From the classical point of view power system instability can be seen as loss of synchronism (i.e., some synchronous machines going out of step) when the system is subjected to a particular disturbance. Three type of stability are of concern: Steady state, transient and dynamic stability.

Steady-state Stability:-

Steady-state stability relates to the response of synchronous machine to a gradually increasing load. It is basically concerned with the determination of the upper limit of machine loading without losing synchronism, provided the loading is increased gradually.

Dynamic Stability:-

Dynamic stability involves the response to small disturbances that occur on the system, producing oscillations. The system is said to be dynamically stable if theses oscillations do not acquire more than certain amplitude and die out quickly. If these oscillations continuously grow in amplitude, the system is dynamically unstable. The source of this type of instability is usually an interconnection between control systems.

Transient Stability:-

Transient stability involves the response to large disturbances, which may cause rather large changes in rotor speeds, power angles and power transfers. Transient stability is a fast phenomenon usually evident within a few second.

Power system stability mainly concerned with rotor stability analysis. For this various assumptions needed such as:

- For stability analysis balanced three phase system and balanced disturbances are considered.
- Deviations of machine frequencies from synchronous frequency are small.
- During short circuit in generator, dc offset and high frequency current are present. But for analysis of stability, theses are neglected.
- Network and impedance loads are at steady state. Hence voltages, currents and powers can be computed from power flow equation.

<u>Dynamics of a Synchronous Machine</u> :-

The kinetic energy of the rotor in synchronous machine is given as:

$$KE=1/2Jw_s^2 x10^{-6} \text{ MJoule}\ldots\ldots\ldots

From equation (5), we can write,

$$M = \frac{2GH}{w_{se}} = \frac{2GH}{2\pi f} = \frac{GH}{\pi f} \text{ MJ.sec/elect. radian} \ldots\ldots\ldots\ldots\ldots\ldots (6)$$

or $$M = \frac{GH}{180f} \text{ MJ.sec/elect. degree} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (7)$$

M is also called the inertia constant.

Assuming G as base, the inertia constant in per unit is

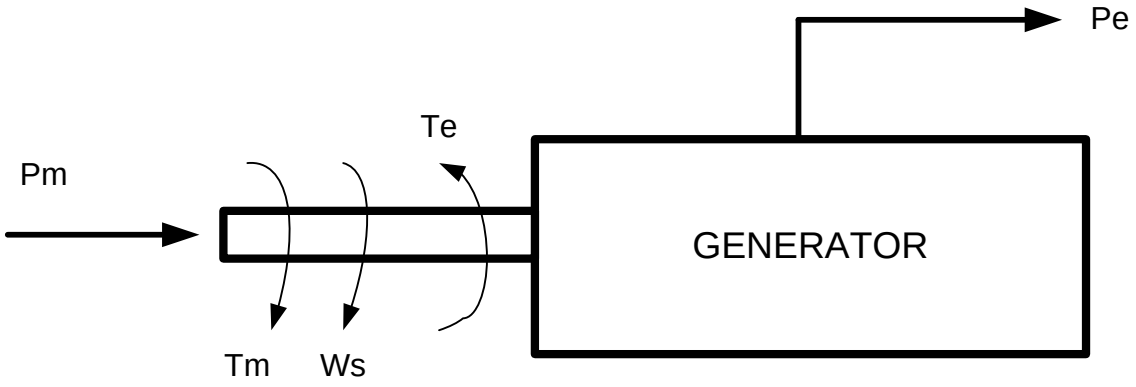
$$M(pu) = \frac{H}{\pi f} \text{ Sec}^2\text{/elect.radian} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (8)$$

or $$M(pu) = \frac{H}{180f} \text{ Sec}^2\text{/elect.degree} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (9)$$

LESSON SUMMARY-2:-

1. Swing equation
2. Multi machine system
3. Machines swinging in unison or coherently
4. Examples

Swing Equation:-



(Fig.-1 Flow of power in a synchronous generator)

Consider a synchronous generator developing an electromagnetic torque T_e(and a corresponding electromagnetic power P_e) while operating at the synchronous speed w_s. If the input torque provided by the prime mover, at the generator shaft is T_i, then under steady state conditions (i.e., without any disturbance).

$$T_e = T_i \text{ ……..} (10)$$

Here we have neglected any retarding torque due to rotational losses. Therefore we have

$$T_e w_s = T_i w_s \text{ ……..} (11)$$

And $$T_e w_s - T_i w_s = P_i - P_e = 0 \text{ ……...……} (12)$$

When a change in load or a fault occurs, then input power Pi is not equal to Pe. Therefore left side of equation is not zero and an accelerating torque comes into play. If Pa is the accelerating (or decelerating) power, then

$$P_i - P_e = M.\frac{d^2\theta_e}{dt^2} + D\frac{d\theta_e}{dt} = P_a \text{ ……...……} (13)$$

Where D = damping coefficient

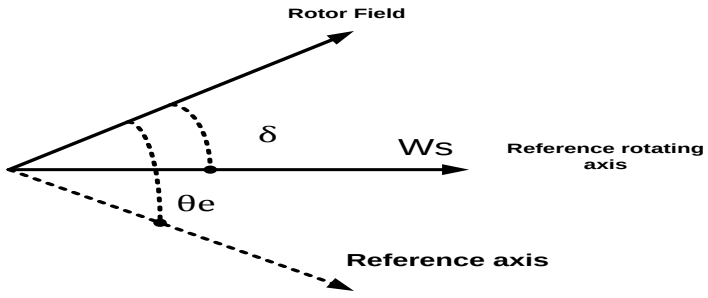
θ_e = electrical angular position of the rotor

It is more convenient to measure the angular position of the rotor with respect to a synchronously rotating frame of reference. Let

$$\delta = \theta_e - w_s.t \text{ ……..……} (14)$$

So $$\frac{d^2\theta_e}{dt^2} = \frac{d^2\delta}{dt^2} \text{ ……...……} (15)$$

Where δ is power angle of synchronous machine.



(Fig.2 Angular Position of rotor with respect to reference axis)

Neglecting damping (i.e., D = 0) and substituting equation (15) in equation (13) we get

$$M.\frac{d^2\delta}{dt^2} = P_i - P_e \text{ MW} \ldots\ldots\ldots\

$$\frac{H_2}{\pi f}.\frac{d^2\delta_2}{dt^2} = (P_{i2} - P_{e2}) \text{ ……..……...…. (25)}$$

Since the machines rotor swing in unison,

$$\delta_1 = \delta_2 = \delta \text{ …….……..(26)}$$

Adding equations (24) and (25) and substituting equation (26), we get

$$\frac{H_{eq}}{\pi f}.\frac{d^2\delta}{dt^2} = (P_i - P_e) \text{ ……...… (27)}$$

Where $P_i = P_{i1} + P_{i2}$

$P_e = P_{e1} + P_{e2}$

$H_{eq} = H_1 + H_2$

Equivalent inertia H_{eq} can be expressed as:

$$H_{eq} = \left(\frac{G_{1,machine}}{G_{System}}\right).H_{1,machine} + \left(\frac{G_{2,machine}}{G_{system}}\right).H_{2,machine} \text{ …….. (28)}$$

Example1:-

A 60 Hz, 4 pole turbo-generator rated 100MVA, 13.8 KV has inertia constant of 10 MJ/MVA.

(a) Find stored energy in the rotor at synchronous speed.
(b) If the input to the generator is suddenly raised to 60 MW for an electrical load of 50 MW, find rotor acceleration.
(c) If the rotor acceleration calculated in part (b) is maintained for 12 cycles, find the change in torque angle and rotor speed in rpm at the end of this period.
(d) Another generator 150 MVA, having inertia constant 4 MJ/MVA is put in parallel with above generator. Find the inertia constant for the equivalent generator on a base 50 MVA.

Solution:-

(a) Stored energy = GH
= 100MVA x 10MJ/MVA
= 1000MJ

(b) $P_a = P_i - P_e = 60-50 = 10MW$

We know, $M = \frac{GH}{180f} = \frac{100X10}{180X60} = \frac{5}{54}$ MJ.sec/elect.deg.\

Now $M.\frac{d^2\delta}{dt^2} = P_i - P_e = P_a$

$\Rightarrow \frac{5}{54}\frac{d^2\delta}{dt^2} = 10$

$\Rightarrow \frac{d^2\delta}{dt^2} = \frac{10X54}{5} = 108$ elect.deg./sec^2

So, α = acceleration = 108 elect.deg./sec^2

(c) 12 cycles = 12/60 = 0.2sec.

Change in $\delta = ½\ \alpha.(\Delta t)^2 = ½.108.(0.2)^2 = 2.16$ elect.deg

Now α = 108 elect.deg./sec^2

= 60 x (108/360°) rpm/sec

= 18 rpm/sec

Hence rotor speed at the end of 12 cycles

$= \frac{120f}{P} + \alpha.\Delta t$

$= (\frac{120X60}{4} + 18X0.2)$ rpm

= 1803.6 rpm.

(d) $H_{eq} = \frac{H_1G_1}{G_b} + \frac{H_2G_2}{G_b} = \frac{10x100}{50} + \frac{4x150}{50}$ =32MJ/MVA

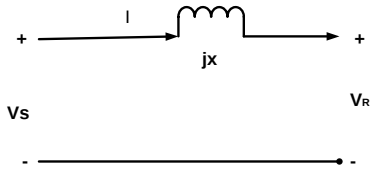
LESSON SUMMARY-3:-

1. Power flow under steady state
2. Steady-state Stability
3. Examples

Power Flow under Steady State:-

Consider a short transmission line with negligible resistance.

V_S = per phase sending end voltage
V_R = per phase receiving end voltage
V_s leads V_R by an angle δ
x = reactance of per transmission line



(Fig.3-A short transmission line)